

## **Exp No: 5**

### **ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS**

#### **AIM:**

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

#### **PROGRAM:**

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;

void setup() {
    //configure pin2 as an input and enable the internal pull-up resistor
    pinMode(12, INPUT_PULLUP);
    pinMode(1,OUTPUT);
    pinMode(2,OUTPUT);
    pinMode(3,OUTPUT);
    pinMode(4,OUTPUT);
    pinMode(5,OUTPUT);
    pinMode(6,OUTPUT);
    pinMode(7,OUTPUT);

}

void loop() {
    //read the pushbutton value into a variable
    int sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {  
  x0=true;  
}  
while(x0){  
  zero();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) {  
    x1=true;  
    x0=false;  
  }  
}  
while(x1){  
  one();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) {  
    x2=true;  
    x1=false;  
  }  
}  
while(x2){  
  two();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x3=true;  
  x2=false;  
}  
}  
while(x3){  
  three();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x4=true;  
  x3=false;
```

```
}  
}  
while(x4){  
  four();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x5=true;  
  x4=false;  
}  
}  
while(x5){  
  five();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x6=true;  
  x5=false;  
}  
}  
while(x6){  
  six();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x7=true;  
  x6=false;  
}  
}  
while(x7){  
  seven();  
  sensorVal = digitalRead(12); if  
(sensorVal == LOW) { x8=true;  
  x7=false;  
}  
}  
while(x8){
```

```

eight();

sensorVal = digitalRead(12); if
(sensorVal == LOW) { x9=true;
x8=false;
}
}

while(x9){
nine();

sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}

}

```

```

void zero()
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
}

```

```

void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
}

```

```
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
  
}  
void two()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,LOW);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,LOW);  
    digitalWrite(7,HIGH);  
delay(delay_time); }
```

```
void three()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
    digitalWrite(7,HIGH);  
delay(delay_time); }
```

```
void four()  
{
```

```
digitalWrite(1,LOW);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void five()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
    digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time); }
```

```
void six()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);
```

```
}  
void seven()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
    digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time); }
```

```
void eight()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void nine()  
{  
    digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);
```

```
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

